HomeRecover Scan

Detailed Hackathon Build Plan

AI-powered post-discharge home safety mapper for the first 48 hours of recovery

Prepared for: CBC McGill Hackathon 2026 — Biology & Physical Health track

Purpose: define the product, technical architecture, phone scanning workflow, deployment path, and execution plan end to end

Core idea: scan a real room/home with a phone, detect recovery hazards, render them in 3D, and generate a condition-specific first-night and first-48-hours safety plan

Plan summary	Decision
Primary demo form factor	Native iPhone app for scanning plus lightweight cloud/web backend for analysis and visualization
Why native iPhone	Apple RoomPlan and ARKit provide the cleanest room capture path on LiDAR-enabled iPhones/iPads; this is the highest-wow and most reliable scanning path
Fallback for non-LiDAR phones	Camera-only guided capture with photo/video upload, then server-side hazard analysis and 2D/3D approximate visualization
MVP boundary	One recovery profile, one route, one or two rooms, 5–7 hazard classes, top 5 interventions, exportable 48-hour checklist
Demo story	Older adult returning home after a fall, now using a walker; scan bedroom to bathroom path; show hazards, safer rearrangement, and caregiver action list

1. Executive Summary

HomeRecover Scan is not a generic “home safety app.” It is a post-discharge recovery support tool. The user scans a bedroom, bathroom, hallway, or small home zone with a phone. The system reconstructs the environment, detects condition-relevant hazards, and generates a personalized 48-hour safety plan for a patient who has just gone home after a fall, fracture, surgery, dizziness episode, C-section, or similar physical-health event.

The strongest hackathon version is centered on a narrow, legible use case: an older adult discharged after a fall and now using a walker. That persona makes the risk logic obvious, the ethical framing strong, and the demo memorable. It also fits the track cleanly because the track explicitly wants tools that help people access care, understand their health, or navigate complex medical systems, including accessibility tools for people with disabilities navigating healthcare.

This plan uses a hybrid architecture. The phone handles scanning and capture. The backend handles hazard interpretation, recommendations, storage, and synchronization. The user-facing result is a room map, a hazard heatmap, a “ghost rearrangement” suggestion layer, and a practical first-night/first-48-hours checklist for the patient or caregiver.

2. Why This Fits the Biology & Physical Health Track

The product is strongest when framed as a care-transition tool, not an AR toy. The real problem is that patients are discharged with restrictions such as “avoid falls,” “limit bending,” “use your walker,” or “watch the wound,” but nobody checks whether the actual home is set up for safe recovery.

This directly supports the track by helping people navigate a confusing part of the medical system: going from clinician instructions to safe real-world implementation at home. It also maps naturally to the track’s example categories of patient education, accessibility, and health literacy for treatment decisions.

The app does not diagnose disease or replace a clinician. It interprets the home environment against a recovery profile and makes environmental recommendations. That is ethically safer and more aligned with the judging rubric than a diagnosis chatbot.

3. Primary User, Persona, and Demo Story

Primary user for hackathon demo: caregiver preparing the home for an older adult discharged after a fall and now using a walker.

Secondary supported users: patient on crutches after fracture, patient after abdominal surgery or C-section with lifting and bending restrictions, patient with dizziness and nighttime fall risk, patient with a wound-care setup need, or anyone with temporary mobility limitation after discharge.

Demo narrative: a caregiver scans the bedroom, hallway, and bathroom before the patient comes home. The app highlights rug risk, narrow turning space for the walker, poor bed-to-bathroom route, and missing reachable essentials. It then proposes a safer path, moved items, and a first-night recovery checklist.

4. Product Definition and Core User Value

The core user value is not “the phone sees the room.” The core value is recovery-specific hazard interpretation. The same object can mean different things for different patients. A coffee table may be harmless for a healthy user, a turning obstacle for a walker user, and a twisting hazard for someone who must avoid bending or pivoting.

The product therefore combines four layers: room capture, object and path understanding, recovery-rule interpretation, and action generation. The action generation is what makes the product useful: it produces concrete interventions before the first night home.

The key output is a prioritized action plan: what to remove, what to move, what route to use, what setup to prepare, and what limitations still require professional guidance.

5. Must-Have Features for MVP

Guided mobile room scan for one or two rooms.

Recovery profile selection: walker after fall or fracture is the recommended MVP profile.

Hazard detection for 5–7 hazard classes.

Simple 3D or pseudo-3D room viewer with highlighted hazards.

Safe-route overlay from bed to bathroom or chair to bathroom.

Ghost rearrangement suggestions for a few obvious items.

First-night and first-48-hours checklist for caregiver or patient.

Clear limitation notice and medical handoff guidance.

6. Stretch Features if Time Allows

Support additional recovery profiles such as crutches, post C-section, or dizziness.

Night mode visualization that highlights low-light bathroom routes.

Discharge-instruction ingestion to personalize rule weighting.

Shareable caregiver link or printable prep sheet.

Multi-room stitching and home-level risk score.

Persistent before/after comparison after rearrangement.

7. Required Hackathon Questions — Ready Answers

Question	Planned answer
Who are we building for, and why do they need it?	Patients and caregivers during the vulnerable first 48 hours after discharge home from a mobility-limiting event. They need help turning vague discharge restrictions into a safe, usable home environment.
What could go wrong, and what would we do about it?	The app could miss hazards, be overtrusted, or collect sensitive home imagery. We mitigate this with conservative wording, local-first capture when possible, manual hazard marking, explicit limitation banners, and human-care handoff guidance.
How does this help people rather than make decisions for them?	The app highlights environmental risks and suggests actions. It does not diagnose, clear the patient for activity, or replace clinicians. The human remains in control.

8. Technical Architecture Overview

The architecture should be intentionally hybrid. Use a native iPhone app for scanning because LiDAR-enabled room capture on Apple devices is best accessed through Apple's native RoomPlan and ARKit stack. Use a backend for hazard analysis, profile logic, storage, analytics, and web-friendly visualization or exports.

A practical stack for a hackathon is: Swift / SwiftUI + RoomPlan / ARKit on iPhone for capture; FastAPI or Node.js backend for processing; Python computer-vision services where needed; Supabase or Firebase for auth/storage; and a simple web dashboard in Next.js or React for reviewing results on laptop or projecting during demo.

Layer	Recommended tech	Purpose
Mobile capture app	SwiftUI + RoomPlan + ARKit + RealityKit	Run guided room scan on LiDAR-enabled iPhone/iPad, preview scan, label room, upload assets
Web dashboard	Next.js or React	Review scan outputs, show checklist, project demo, optional caregiver share screen
API/backend	FastAPI or Node.js/Express	Receive uploads, process scan metadata, run rules engine, create recommendations, persist session data
CV / hazard service	Python + OpenCV / ML helper models	Detect hazard classes from images or scan metadata, estimate routes, classify clutter and floor obstacles
Database + storage	Supabase or Firebase Storage	Store user sessions, room JSON, annotations, screenshots, and generated plans
Visualization	RealityKit on device, and/or Three.js on web	Render highlighted rooms, paths, and ghost rearrangement overlays
LLM layer	Claude API as explanation layer only	Turn structured hazards into natural-language guidance and checklist copy without making the LLM the core product

9. How Phone Scanning Should Work

There are two scanning modes. The primary path is LiDAR-capable iPhone/iPad scanning. The fallback path is camera-only guided capture for non-LiDAR devices. For the hackathon, optimize hard for the primary path and make the fallback basic but presentable.

The scan experience should be intentionally short and narrow. For MVP, do not attempt an entire house. Limit the capture to the recovery-critical zone: bedroom, hallway, bathroom, or a single route such as bed to bathroom.

9.1 Primary path: LiDAR-enabled iPhone/iPad

Use Apple RoomPlan to capture room geometry and major furniture/object categories. RoomPlan is designed for creating a 3D floor plan of a room using the camera and LiDAR scanner on supported iPhone

and iPad Pro devices. This gives you a far better “wow” factor and a cleaner engineering story than trying to fake a room mesh from scratch in a browser.

During scan, the app should show simple guidance: move slowly, keep the room well lit, capture the whole path, and pause briefly around the bed, chair, bathroom threshold, rug area, or other high-risk objects. Mirrors, glass, very dark surfaces, and poor lighting can degrade scan quality, so the interface should warn users before they begin.

When scan completes, save three outputs: RoomPlan / ARKit room metadata, preview images or screenshots, and your own lightweight room-session JSON that maps scan objects to your internal risk engine.

- Minimum viable scan loop: choose profile → choose room type → run guided scan → preview hazards → confirm upload
- For demo day, prepare one known-good room and one known-good LiDAR device so the live scan is reliable
- Also capture a prerecorded scan session in case the venue lighting or network is poor

9.2 Fallback path: non-LiDAR phone

If a device does not have LiDAR, do not promise the same fidelity. Instead, present a fallback “guided capture” mode: the user takes 6–10 photos or a short slow video of the room and the critical route. The backend then performs hazard detection and creates a 2D floor-path approximation or a simplified pseudo-3D scene.

This fallback is useful for product completeness and inclusion, but it should not be the centerpiece of the hackathon demo. The best hackathon story is still the LiDAR path, because it is much more defensible and visually stronger.

Question	Recommended answer
Will LiDAR be used?	Yes — for the strongest version, LiDAR-enabled iPhones/iPads should be used for the primary room-scanning flow.
Do we need LiDAR to have a product?	No — but the fallback should be treated as reduced-fidelity guided capture, not equal-fidelity replacement.
Can this be pure browser-based on iPhone?	Not for the full LiDAR / RoomPlan experience. The clean path is a native iOS capture app plus web/backend layers around it.
Should we still have a web layer?	Yes — for review, projection, exports, and optional caregiver sharing.

10. Hazard Taxonomy and Rules Engine

The rules engine is the brain of the product. The scan itself is only useful if it is translated into recovery-specific risk. Keep this engine explicit and explainable. Do not hide the entire decision process inside an LLM.

For MVP, use a hand-built rules engine with weighted scores rather than a black-box model. Judges can understand it, and you can defend every recommendation.

Hazard class	Examples	Why it matters for walker-after-fall profile
Floor obstacle / trip risk	Loose rugs, cords, shoes, clutter piles	Increases fall risk, especially during turns or at night
Path-width obstruction	Coffee tables, chair placement, narrow doorway approach	Reduces turning radius and safe walker navigation
Transfer challenge	Low bed, low couch, no nearby stable surface	Makes sit-to-stand and stand-to-sit more dangerous
Bathroom risk	Wet floor zone, no easy support surface, awkward threshold	Bathroom is a high-risk fall environment and often the first-night problem zone
Reachability issue	Essentials too high, too low, or too far from bed/chair	Forces bending, twisting, or unstable walking
Lighting / night-route issue	Dark path to bathroom, no lamp near bed	Nighttime increases misstep risk
Wound-care / hygiene setup issue	No clean surface, poor placement of supplies	Creates unsafe or impractical recovery routines for some profiles

10.1 Example rule logic

- If profile = walker_after_fall and object = rug on primary path, then mark urgent hazard. Recommendation: remove or secure rug before first night home.
- If profile = walker_after_fall and bed_to_bathroom path width is below threshold in one or more segments, then mark urgent path restriction. Recommendation: move chair or table to widen route.
- If profile includes no_bending or no_heavy_lifting and essentials are stored low, then mark reachability issue. Recommendation: elevate medications, water, charger, and wound-care items to waist height.
- If profile includes dizziness and route lighting is low, then mark night-route hazard. Recommendation: add bedside lamp, motion light, or illuminated bathroom route.

11. User Experience Flow

Flow step	What happens
Step 1 — Start session	User opens app, selects “Prepare home for recovery,” and chooses recovery profile.
Step 2 — Add constraints	User indicates walker, cane, crutches, no bending, dizziness, stairs, lives alone, caregiver support, etc.
Step 3 — Scan room or route	Native app runs RoomPlan capture. UI guides user to scan bed, bathroom entrance, floor path, and key furniture.
Step 4 — Preview detected environment	Show basic room map with identified furniture and hazard pins.
Step 5 — Analyze	Backend or on-device rules engine scores hazards and route risk.
Step 6 — Show results	Display 3D room with red/yellow markers, safe path overlay, and ghost rearrangement suggestions.

Flow step	What happens
Step 7 — Action plan	Generate top 5 fixes before patient arrives, first-night instructions, and 48-hour checklist.
Step 8 — Share/export	Optionally send to caregiver, save PDF, or show on web dashboard for demo.

12. 3D Visualization Plan

The 3D experience should be impressive but constrained. It is not worth spending the hackathon building a fully polished interior-design engine. Build exactly enough 3D to make the risk interpretation visible and memorable.

Three 3D elements matter most: hazard heatmap, safe path overlay, and ghost rearrangement suggestions. If those are clear, the visualization already feels advanced.

- Hazard heatmap: color-coded object and floor-region overlays for urgent, moderate, and low-risk findings
- Safe route: draw the preferred path from bed to bathroom or chair to bathroom
- Ghost rearrangement mode: show translucent moved furniture or removed rug suggestions so the user can see the safer layout
- Reachability mode: optional highlight for items that should be within comfortable reach from bed or chair
- Night mode: optional darkened scene emphasizing hazards along nighttime walking path

13. How to Deploy So It Works on Phone While You Build on Laptops

This is the practical workflow. You will write most of the code on your laptops in VS Code or your usual editor, but the actual iPhone scanning app must be built and signed through Xcode on a Mac because the LiDAR and RoomPlan path is native iOS.

Think of the project as two deliverables built in parallel: a native capture client and a backend/web system. Most of the team can stay in VS Code working on backend, rules, web visualization, API routes, and storage. One person needs to own the Swift / Xcode capture app.

Component	Where you build it	How you run/test it
iPhone scan app	Xcode on Mac (SwiftUI / RoomPlan / ARKit)	Run directly on your iPhone via cable or local developer signing; if needed, distribute beta via TestFlight
Backend API	VS Code on laptop	Run locally during dev; expose to phone with ngrok/cloudflared tunnel, or deploy to a cloud host
Web dashboard	VS Code on laptop	Run local dev server on laptop and access from browser, or deploy to Vercel/Netlify for easy demo
Database/storage	Cloud dashboard + code in VS Code	Use hosted Supabase/Firebase so phone and laptop talk to same source of truth

Component	Where you build it	How you run/test it
Shared assets	Local repo / GitHub	Use GitHub for source, issues, and handoff between native and backend teammates

13.1 Recommended dev/test setup

Fastest hackathon setup: deploy backend and storage to the cloud, then point the iPhone app to those URLs. This avoids LAN/network weirdness on event day.

If you are still developing locally, use a tunneling service such as ngrok or Cloudflare Tunnel to expose your local API to the phone. That lets the phone upload scan sessions directly to your laptop-hosted backend while you iterate.

For demo stability, have a production-like deployed backend before hackathon day. The last thing you want is a laptop firewall issue or unstable tunnel during judging.

14. LiDAR-Specific Plan

Item	Plan
Primary supported hardware	LiDAR-enabled iPhone Pro models or supported iPad Pro devices for RoomPlan-based scanning
Environment guidance	Well-lit room, slow scan motion, minimal glass/mirror confusion, avoid scanning too much irrelevant space
Asset outputs	Room geometry / object categories from RoomPlan, preview snapshots, session metadata JSON, optional exported mesh or floor-plan data
Demo risk mitigation	Bring a supported device, pre-scan a room beforehand, and store a backup completed session to show even if live scan fails
If no LiDAR device is available	Use fallback guided capture path and present 3D as reduced-fidelity reconstruction rather than LiDAR-grade room mapping

15. Data Flow and Integration Sequence

- 1. User opens app on iPhone and selects recovery profile.
- 2. Phone captures room data with RoomPlan / ARKit and stores temporary local session state.
- 3. User confirms upload. Phone sends room metadata, screenshots, profile parameters, and optional room labels to backend API.
- 4. Backend normalizes raw capture outputs into internal spatial objects and route graph.
- 5. Hazard engine scores hazards based on selected profile and room structure.
- 6. Recommendation engine generates prioritized actions, safe-path suggestions, and ghost rearrangement targets.
- 7. Claude API, if used, converts structured findings into concise caregiver-facing wording and checklist language.
- 8. Backend persists results to database/storage and returns a session result bundle.

- 9. Mobile app renders immediate summary; web dashboard can fetch same session for projection and export.
- 10. Optional export step creates shareable PDF checklist for caregiver or patient.

16. Detailed Build Plan by Workstream

Workstream	Owner focus	Deliverables
Native capture	Swift/iOS teammate	Scan screen, room labeling, upload client, local preview, minimal RealityKit visualization
Backend + rules	Python or Node teammate	API routes, hazard scoring engine, recommendation generation, profile schema
Web/demo UI	Frontend teammate	Session review page, 3D or pseudo-3D display, checklist panel, share/export page
Product/design	Any teammate / lead	Prompt flows, copy, icons, risk wording, ethical warnings, demo narrative
Data/demo prep	Any teammate	Prepare sample room, prerecorded scan, screenshots, fixture JSON, fallback assets

16.1 Day-by-day execution plan

Time block	Primary objective	Concrete tasks
Pre-build Day 1	Lock scope and scaffolding	Choose single persona and route; create repo; scaffold Swift app, backend, storage, and web dashboard; define JSON schemas and hazard classes
Pre-build Day 2	End-to-end scan loop	Get room scan working on phone; upload session to backend; persist session; show first result page on web
Pre-build Day 3	Risk engine + first wow layer	Implement hazard scoring for 5–7 classes; add safe path and at least one ghost rearrangement feature; create first-night checklist
Hackathon morning	Polish and reliability	Tune UI, fix bugs, preload demo data, refine wording, add limitation banners, prepare presentation
Before submission	Judging readiness	Freeze a stable build, export screenshots/video, confirm backup demo path, rehearse 5-minute pitch twice

16.2 Task breakdown in finer detail

Native iOS capture app

- Create SwiftUI app shell with onboarding and profile selection.
- Integrate RoomPlan capture view.
- Add room-type label and route label (for example, bedroom or bathroom).
- Persist local session bundle and upload to backend.
- Add basic session preview page with success/failure handling.
- Handle scan cancellation, poor scan quality, and retry states.

Backend and scoring

- Define profile schema, hazard schema, and recommendation schema.
- Write endpoint to receive scan bundle.
- Normalize RoomPlan outputs into internal objects and floor/path primitives.
- Implement rule engine with weighted severity and recommendation templates.
- Generate result payload for mobile and web clients.
- Store everything in hosted database and object storage.

Visualization and demo

- Build session review page.
- Add hazard list sorted by severity.
- Add 3D viewer or 2.5D room map with safe-path overlay.
- Add ghost rearrangement toggle.
- Add first-night and 48-hour checklist panels.
- Add export button and shareable screenshots.

Ethics and copy

- Write limitation banner.
- Add handoff language for “contact your care team” scenarios.
- Make sure no copy implies diagnosis, clearance, or medical replacement.
- Provide privacy statement for captured room data.
- Add “manual mark hazard” tool if a hazard is missed.

17. Suggested API Design

Endpoint	Method	Purpose
/sessions	POST	Create scan session and return upload/session IDs
/sessions/{id}/upload	POST	Upload room metadata, screenshots, optional mesh, and profile parameters
/sessions/{id}/analyze	POST	Run hazard scoring and recommendation generation
/sessions/{id}	GET	Fetch full analyzed session bundle
/sessions/{id}/export	GET	Return printable checklist or PDF summary
/profiles	GET	Return supported recovery profiles and parameter defaults

18. Suggested Data Model

Entity	Key fields
Session	session_id, created_at, user/device, recovery_profile, scan_mode, status
Room	room_id, session_id, room_type, dimensions, route_labels, scan_quality
Object	object_id, room_id, category, position, dimensions, confidence, on_primary_path
Hazard	hazard_id, session_id, class, severity, explanation, related_object_ids, recommendation_ids
Recommendation	recommendation_id, session_id, priority, category, text, target_location, expected_benefit
Export	export_id, session_id, pdf_url, screenshot_urls, share_token

19. Privacy, Safety, and Ethical Guardrails

Because this product captures highly personal home imagery and is adjacent to health decision-making, the ethical layer needs to be explicit. This is not optional window dressing for the hackathon; it is a scoring advantage.

The product should present itself as environmental recovery support, not medical clearance. The app should state that it may miss hazards, that it is not a substitute for clinician, occupational therapist, or physiotherapist guidance, and that urgent symptoms require direct medical care.

Risk	Mitigation
Missed hazard / false negative	Manual hazard marking, conservative language, backup checklist, explicit “this may miss issues” notice
Overreliance on the app	Clear banner: environmental guidance only, not diagnosis or medical clearance
Privacy of room scans	Minimize retention, secure storage, delete test scans, avoid collecting more imagery than needed
Exclusion of users without LiDAR	Provide fallback capture mode and caregiver-assisted scan option
Bias or poor fit for some homes/users	Keep rules visible and editable where possible; allow manual context notes

20. Judging Strategy

Optimize for the rubric in order: technical execution, real-world impact, ethical alignment, and presentation quality. The project will stand out if the judges can instantly see that this is not “ChatGPT but for health.” It

is a physically grounded, post-discharge workflow tool with computer vision, spatial reasoning, and a constrained clinical role.

For technical questions, be ready to explain exactly why a native capture app was used, how room data becomes hazards, why the rules engine is explicit, where the LLM is and is not used, and what the backup plan is if live scanning fails.

- Show the live or prerecorded scan first — get the wow moment early
- Immediately tie visuals to a vulnerable patient persona
- Show actionable output, not just pretty 3D
- Say clearly what the app does not do
- End on why this matters in the first 48 hours after discharge

21. 5-Minute Demo Script

Time	What to show/say
0:00–0:30	Open with the care-transition problem: discharge instructions are vague, but the home environment determines safety.
0:30–1:00	Introduce persona: older adult coming home after a fall, now using a walker.
1:00–2:00	Run or show phone scan of bedroom to bathroom route.
2:00–3:00	Reveal hazard map and explain top three hazards in plain language.
3:00–4:00	Toggle ghost rearrangement and safe path overlay; show first-night checklist.
4:00–4:30	Show limitation banner and explain handoff role — supports recovery, does not replace clinicians.
4:30–5:00	Close with impact: safer first night home, clearer caregiver prep, better discharge follow-through.

22. Failure Modes and Backup Plans

Failure mode	Backup plan
Live scan misbehaves	Use a prerecorded scan session and continue with analyzed results
Venue internet is unstable	Cache one complete local demo bundle and keep screenshots/video locally on laptop and phone
LiDAR device unavailable	Switch to fallback capture demo and be explicit that this is reduced fidelity
3D renderer breaks	Use static screenshots plus hazard list and path overlay on 2D floor map
LLM latency	Keep structured recommendations ready; LLM wording should be optional polish, not a hard dependency

23. Final Recommendation

Build this as a native iPhone LiDAR-first capture experience with a cloud backend and a web review layer. That is the cleanest path technically and the strongest story for judging. Most of your code can still live in VS Code on your laptops, but the scanning client itself should be implemented in Swift/Xcode on a Mac and run on a real iPhone for the demo.

Keep the MVP brutally focused: one persona, one route, one or two rooms, 5–7 hazard classes, top five fixes, first-night checklist, and one wow visualization layer. That is enough to feel deep, differentiated, and responsibly aligned with the Biology & Physical Health track.

Do not let the 3D visuals eat the project. The visual layer is there to make the action plan legible. The actual product is a post-discharge environmental risk engine for safer recovery at home.

24. Suggested Repository Structure

Path	Purpose
ios-app/	SwiftUI, RoomPlan, ARKit, capture screens, upload client, local preview
web-dashboard/	React or Next.js session viewer, checklist page, optional share/export UI
backend/	FastAPI or Node service, auth, session ingestion, analysis orchestration
backend/rules/	Recovery profiles, severity weights, hazard templates, recommendation logic
backend/tests/	Fixture scans, rule-engine tests, sample analyzed sessions
storage-fixtures/	Known-good screenshots, prerecorded scan bundles, pitch assets
docs/	Architecture diagram, API schema, demo script, judging notes

25. Demo-Day Readiness Checklist

- Bring at least one supported LiDAR iPhone or iPad with the app already installed and signed.
- Keep one completed, analyzed session offline on both the phone and a laptop.
- Store screenshots of hazard map, safe path, ghost rearrangement, and checklist for instant fallback.
- Verify backend URL, auth keys, storage bucket permissions, and export route before leaving for the event.
- Rehearse answers to: why native iPhone, why not just an LLM, where LiDAR is used, and what happens when scan quality is poor.
- Prepare a one-line ethical disclaimer and a one-line impact statement for the final 20 seconds of the pitch.

Source note: This plan was aligned to the official CBC McGill Hackathon 2026 brief for the Biology & Physical Health track and its judging and ethics expectations, plus current Apple platform guidance for RoomPlan, ARKit, and TestFlight-based native iOS testing/distribution.